Theoretical Feasibility of Circumventing Hypervector Capacity Limits via Neurogenesis-Inspired Whitening Algorithms

Introduction to Hyperdimensional Computing and the Capacity Crisis

Hyperdimensional Computing (HDC), formally encompassed within the broader theoretical framework of Vector Symbolic Architectures (VSA), represents a fundamental paradigm shift in computational cognitive modeling, artificial intelligence, and neuromorphic engineering.[1] Deviating from traditional von Neumann computing and localized neural network architectures, HDC draws profound inspiration from the high-dimensional, highly distributed, and holographic nature of neural circuits in the biological brain.[3] Within this paradigm, information is not stored in precise, localized floating-point scalars, but is instead holistically distributed across massive, fixed-length vectors—termed hypervectors—typically operating in spaces ranging from 1,000 to 10,000 dimensions.[2]

The primary advantage of this holographic representation is its intrinsic robustness to noise, hardware failures, and data uncertainty; because no single component of a hypervector is uniquely responsible for a specific piece of information, the degradation of individual components does not result in catastrophic memory loss.[6] HDC models, including Kanerva's Sparse Distributed Memory (SDM) and Holographic Reduced Representations (HRR), utilize a complete algebraic system of binding, bundling, and permutation to encode highly complex hierarchical data structures, sequences, and semantic relationships into a single, uniform vector space.[1]

However, the mathematical elegance of HDC is inherently constrained by a strict mathematical bottleneck: the capacity limit of superposition.[9] As multiple hypervectors are continuously bundled (superposed) to represent an expanding set of features, memories, or classes, the resulting composite vector gradually loses its specific orthogonality relative to its constituent elements.[9] The signal-to-noise ratio (SNR) decays exponentially with each addition, generating a dense entanglement of data that eventually reaches a saturation point where original constituents can no longer be reliably retrieved.[10]

To bypass these fundamental capacity bottlenecks, an emerging theoretical framework proposes the integration of mechanisms derived from biological adult neurogenesis—the process by which the mature brain dynamically generates new neurons to accommodate novel

information without catastrophically overriding existing memories.[12] By fusing the biological principles of structural plasticity with unsupervised Hebbian learning rules—specifically Oja's rule and Sanger's Generalized Hebbian Algorithm (GHA)—it becomes theoretically feasible to implement a continuous, online "whitening" process.[14] This proposed algorithmic mechanism decorrelates saturated representations and dynamically grows new, orthogonal vector dimensions when a critical capacity threshold—operationally defined across multiple domains as the 75 percent capacity threshold—is approached.[16] The resulting neurogenesis-driven algorithm would effectively transform HDC from a static, fixed-dimensional vector space into an adaptive, lifelong-learning architecture capable of boundless capacity expansion.

The Mathematical Mechanics of Vector Symbolic Architectures

To accurately model a solution to the hypervector capacity limit, it is first necessary to dissect the algebraic operations of Vector Symbolic Architectures and the exact nature of their failure modes. In standard binary or bipolar HDC systems, cognitive information processing is executed via three primary operations, which together form a ring-like algebraic structure over the high-dimensional space.[6]

HDC Operation	Mathematical Implementation (Bipolar)	Primary Function	Impact on Vector Capacity
Binding	Element-wise multiplication ($\otimes$)	Associates variables with values; creates hierarchical data structures.	Generates a new vector orthogonal to operands; neutral to superposition limits.
Permutation	Cyclic coordinate shift (ρ)	Encodes sequential order and temporal dynamics without commutative ambiguity.	Preserves orthogonality and dimensionality; neutral to superposition limits.
Bundling	Element-wise addition + Thresholding ($\oplus$)	Combines multiple vectors into a composite set memory.	**Reduces SNR; drives the system toward capacity saturation and crosstalk.**

The operation responsible for the capacity crisis is bundling. When a system bundles a set of hypervectors to form a class representation, the composite hypervector must maintain a sufficiently high cosine similarity to each stored vector to allow for accurate nearest-neighbor retrieval.[9] Because hypervectors are drawn from an independent and identically distributed (i.i.d.) random space, any two randomly chosen vectors are nearly orthogonal (i.e., their cosine similarity approaches zero).[10]

However, as the number of bundled vectors n increases, the variance of the composite vector expands. The relative distance between the composite vector and any individual constituent shrinks toward the distance of two random, orthogonal vectors.[9] This phenomenon manifests as an accumulation of crosstalk. The interference from other stored patterns acts as additive noise during the retrieval process.[20] At a specific mathematical juncture, the SNR falls below the margin of error required for accurate classification, resulting in a state where the HDC model suffers from catastrophic forgetting, unable to distinguish previously learned patterns from random noise.[11]

The 75 Percent Capacity Threshold: A Cross-Disciplinary Convergence

In addressing the limits of hypervector superposition, a heuristic threshold of "75 percent" emerges repeatedly across biological, hardware, and algorithmic domains as a critical inflection point for phase transitions, capacity saturation, and system stress.[16] Rather than being a purely arbitrary metric, the 75 percent capacity threshold serves as a rigorously backed trigger point for initiating structural expansion.

Hardware and Processing-In-Memory (PIM) Constraints

HDC is highly favored for edge computing and embedded devices because its lightweight operations map perfectly to low-power Field-Programmable Gate Arrays (FPGAs) and Processing-In-Memory (PIM) architectures.[7] However, these hardware implementations face strict physical capacity limits. Research into ReRAM-based PIM architectures demonstrates that when the total CMOS circuitry (including lanes, hash decoders, registers, and routers) reaches approximately 75 percent area utilization, the system begins to experience severe diminishing returns in energy efficiency and read-disturb vulnerabilities.[21] PIM crossbars suffer from severe analog noise when densely packed, and pushing utilization past the 75 percent limit induces off-chip memory accesses that entirely negate the high computing efficiency of the ReRAM.[24]

Signal-to-Noise Ratio (SNR) and Data Robustness

From an algorithmic robustness perspective, HDC systems exhibit an extraordinary capacity to

tolerate noise. Studies deploying expeditionary robots trained via HDC demonstrate that the system can solve complex navigation tasks even when up to 75 percent of the sensors feed completely irrelevant data or noise to the robot.[16] However, once the noise floor or crosstalk exceeds this 75 percent threshold, the mathematical pseudo-orthogonality of the hypervectors breaks down, and the system fails catastrophically.[10]

Biological Reservoir Capacities

In biological neural stem cell (NSC) hierarchies, which govern adult neurogenesis, the 75 percent threshold also appears as a critical regulatory boundary. Studies tracking multicellular dynamics and progenitor pools in the brain reveal that the deeply quiescent and self-renewing "reservoir" NSCs (rNSCs) account for approximately 75 percent of the Sox2-expressing stem cell population.[22] The remaining cells exist in an activated, operational state supporting the bulk of ongoing neurogenesis.[22] When the activated pool is depleted or the environmental demand for new memory representations increases, the system must tap into this 75 percent reservoir to sustain neurogenesis.[22]

Therefore, translating this convergence to Hyperdimensional Computing, when a hypervector bundle reaches 75 percent of its theoretical Macrow or Kanerva memory capacity limit, the probability of retrieval errors accelerates exponentially. The 75 percent capacity mark functions as the optimal logical trigger point for adaptive intervention. Instead of allowing the system to reach 100 percent saturation—which would result in unrecoverable crosstalk—the algorithm must dynamically trigger a neurogenesis-like expansion event to preserve previously learned representations.

Biological Precedents: Adult Neurogenesis and Progenitor Triggers

To successfully architect an algorithmic intervention at this 75 percent capacity threshold, neuromorphic engineering must meticulously model the biological brain. The mammalian brain avoids catastrophic forgetting and handles continuous capacity demands through structural plasticity, most notably via adult neurogenesis.[13] Neurogenesis is the generation of new neurons and their integration into existing, mature neural circuits.[12] This process continues throughout adulthood in highly specific regions, such as the dentate gyrus of the hippocampus and the olfactory bulb, providing a lifelong mechanism for adapting to novel environments without overwriting established memory traces.[12]

Transcriptional Networks and Capacity Triggers

The generation of new neurons is not a random process; it is a highly regulated response to environmental complexity and capacity demands.[27] In the developing ventral midbrain, specific radial glia (Rgl) subtypes dictate capacity expansion. The Rgl1 subtype acts as the primary

progenitor for the midbrain dopaminergic (mesDA) lineage.[27] Gene Set Enrichment Analysis (GSEA) reveals that Rgl1 is functionally specialized and uniquely enriched in genes related to cell proliferation.[27]

Crucially, the decision for Rgl1 to proliferate and spawn new neurons is triggered by a specific transcriptional network coordinated by the transcription factor BMAL1.[27] BMAL1 serves as a master regulator of mesDA neurogenesis, acting as the biological "switch" that monitors the system's need for new computational nodes.[27]

Furthermore, this proliferation is tightly constrained by the system's metabolic state. The Rgl2 subtype is enriched in genes related to mitochondrial respiration and lipid metabolism, providing the energetic criteria required to support the high metabolic demands of structural growth.[27] Simultaneously, the Rgl3 subtype serves as a "signaling niche," expressing extracellular matrix (ECM) molecules such as Netrin-1 (Ntn1) and Slit1, which promote the survival and yield of newly generated neurons, ensuring they integrate successfully into the existing neural architecture without causing disruptive interference.[27]

In a computational equivalent, an advanced HDC system must possess a "progenitor" monitoring mechanism akin to the Rgl1/BMAL1 complex. This mechanism must continuously evaluate the metabolic state (the hardware memory footprint) and the signaling environment (the SNR and cosine similarity of the current bundled hypervector representation). When the SNR degrades to the 75 percent capacity threshold, the artificial BMAL1 analogue is triggered, initiating the algorithmic generation of a new vector dimension.

The Sparse Updating Hypothesis

Another critical biological concept that directly informs the architecture of a dynamic HDC algorithm is the "sparse updating hypothesis," which is heavily observed in hierarchical linguistic predictions and narrative comprehension within the human brain.[27]

Neurolinguistic research comparing how the brain updates information across multiple timescales—from short phonemes to long sentences—identifies two rival hypotheses:

1. **Continuous Updating Hypothesis:** Posits that higher-level brain regions are updated continuously and gradually as inputs from lower levels unfold over time.[27]
2. **Sparse Updating Hypothesis:** Posits that higher-level updates occur *only* at the boundaries of their preferred timescales (e.g., at the end of a sentence or narrative chunk). This leads to abrupt, discrete changes in neural responses.[27]

Computational modeling and functional magnetic resonance imaging (fMRI) data demonstrate that the brain heavily favors the sparse updating model, particularly in higher-order regions like the Default Mode Network (DMN).[27] Sparse updating allows the brain to maintain representational stability; it accumulates information at a lower level and only triggers a state update in the higher-level architecture when a specific boundary condition is met.[27] This sparse

approach is significantly more computationally efficient and accelerates the updating process.[27]

This biological sparse updating mechanism perfectly dictates the proposed HDC capacity intervention. Continuously adding dimensions to a hypervector space as new data arrives (continuous updating) would be computationally disastrous, completely destroying the fixed-size algebraic properties of VSA.[27] Instead, the system must maintain a static dimensionality until the 75 percent capacity threshold (the structural boundary) is breached.[27] Only at this discrete boundary does the sparse update trigger, executing a neurogenesis-like expansion event.

Neural Population Geometry and the Imperative of Whitening

Before new vector dimensions can be integrated via neurogenesis, the problem of the saturated, highly correlated representation within the existing hypervector must be addressed. Simply appending new random dimensions to a vector that is already suffering from severe crosstalk does not resolve the underlying interference. The geometry of neural populations provides the mathematical foundation for resolving this entanglement.[27]

Generalization performance in neural networks processing tasks with shared latent structures is determined by four key geometric statistics: dimensionality, correlation structures, signal-noise factorization (SNF), and signal-signal factorization (SSF).[27]

Factorized Representations and Coding Efficiency

Factorization, or disentanglement, occurs when independent environmental variables are encoded along distinct, orthogonal directions of variation within the neural activity space.[27] In an ideal HDC system, every newly bound concept would occupy a perfectly orthogonal subspace. However, because real-world data contains highly correlated features, naïve bundling inevitably leads to representations that are highly entangled.[28] As the hypervector bundle approaches the 75 percent capacity limit, the lack of Signal-Signal Factorization (SSF) causes the representation to collapse into a dense, uninterpretable state.[27]

Early in learning, optimal neural representations are lower-dimensional and exhibit high correlations to task variables.[27] However, as the system encounters more complex, multiplexed tasks, coding efficiency demands a shift toward higher dimensionality and strict orthogonalization.[27] To achieve this, the data must undergo a mathematical process known as "whitening."

Whitening as a Mechanism for Capacity Maximization

Whitening (also known as sphering) is a linear transformation that converts a vector of random

variables with a known covariance matrix into a set of new variables whose covariance is the identity matrix.[15] This transformation ensures that the resulting variables are completely uncorrelated and possess a uniform variance of one.[15]

In statistical signal processing and theoretical neuroscience, whitening is the ultimate mathematical tool for maximizing information transmission in the presence of noise.[28] In the context of the olfactory bulb, adult neurogenesis serves precisely this function; the continuous integration of new granule cells allows the cortical representations to whiten odor representations, separating overlapping olfactory signals into distinct, decorrelated channels.[12] Similarly, in human perception models, whitening stimuli removes chance correlations, providing a greater than 85 percent improvement in sampling efficiency and drastically increasing estimation accuracy.[29]

By applying a whitening transformation to a saturated hypervector bundle, the system effectively decorrelates the overlapping patterns. It flattens the eigenvalue spectrum of the data covariance matrix, ensuring that the neural code distributes variance equally across all available hidden units or dimensions.[33] If a bundle has reached the 75 percent capacity threshold due to overlapping, correlated data points, whitening acts as a critical reset mechanism. It compresses the redundant data, separates the signal from the noise (enhancing Signal-Noise Factorization), and extracts the true, independent latent features.[27]

However, standard mathematical whitening requires computing the full covariance matrix of the data and performing an eigenvalue decomposition (EVD) offline—an operation with cubic computational complexity $O(N^3)$.[30] This heavy computational burden is entirely antithetical to the lightweight, highly parallel, streaming nature of Hyperdimensional Computing.[7] To implement whitening dynamically as a precursor to neurogenesis without stalling the system, HDC requires an online, iterative learning rule.

Hebbian Plasticity: Oja's Rule and Sanger's Rule (GHA)

To circumvent the computational bottleneck of offline whitening and Principal Component Analysis (PCA), neuromorphic models rely on biologically plausible Hebbian learning rules. The most prominent and mathematically rigorous of these are Oja's rule and Sanger's rule, which serve as the algorithmic engines for adaptive dimensionality and neurogenesis.[14]

Oja's Rule and Multiplicative Normalization

Standard Hebbian learning, frequently summarized by the axiom "neurons that fire together, wire together," is fundamentally unstable. If synaptic weights are updated purely based on the correlation of pre- and post-synaptic activity, the weights will grow exponentially toward infinity, eventually destabilizing the entire network.[35]

Erkki Oja (1982) resolved this instability by introducing a competitive, homeostatic weight

decay term, resulting in what is universally known as Oja's rule.[35] Mathematically, Oja's rule updates the synaptic weight vector w based on the input x and the scalar output $y = w^T x$ according to the equation:

$$\Delta w = \alpha(xy - y^2 w)$$

where α represents the learning rate.[35] The critical innovation is the term $-y^2 w$, which acts as a subtractive normalization mechanism.[35] This term continuously constrains the norm of the weight vector to unity, preventing indefinite growth.[35] Remarkably, this simple, localized learning rule converges precisely to the first principal component of the input data.[35] Oja's rule allows a single artificial neuron to dynamically discover the direction of maximum variance in a streaming dataset without ever explicitly computing a covariance matrix.[36]

Sanger's Rule: The Generalized Hebbian Algorithm

While Oja's rule successfully finds the first principal component, resolving a hyperdimensional capacity crisis requires finding and separating *multiple* orthogonal components. Terence Sanger (1989) expanded Oja's concept into the Generalized Hebbian Algorithm (GHA), commonly referred to as Sanger's rule.[14] Sanger's rule enables a single-layer feedforward network to extract multiple principal components sequentially and, crucially, orthogonally.[14]

Sanger's rule incorporates a Gram-Schmidt-like orthogonalization process directly into the Hebbian update.[15] For the i-th neuron in the network, the weight update mathematically subtracts the projections of the input onto the previously learned $i - 1$ principal components:

$$\Delta w_{ij} = \eta y_i \left(x_j - \sum_{k=1}^{i} y_k w_{kj} \right)$$

By forcing each subsequent neuron to learn only from the residual variance left over by the preceding neurons, Sanger's rule guarantees that the resulting weight vectors become perfectly orthogonal to one another.[15] The outputs are strictly decorrelated, directly accomplishing the whitening process in an online, streaming fashion.[15]

Sanger's rule is the quintessential algorithm for dynamic representation expansion. Because HDC relies fundamentally on the pseudo-orthogonality of its hypervectors, any newly grown vector dimension must not interfere with the existing space.[14] By utilizing Sanger's rule, the system inherently builds a set of mutually orthogonal basis vectors, satisfying the exact requirement for maintaining high capacity and low crosstalk in Vector Symbolic Architectures.[14]

Pehlevan's Framework: Adaptive Dimensionality and

Algorithmic Neurogenesis

The theoretical bridge connecting Oja/Sanger whitening rules, Principal Component Analysis, and biological neurogenesis was formally established by Cengiz Pehlevan and Dmitri Chklovskii.[28] Their research derived online PCA and whitening algorithms from a "similarity matching" framework, utilizing both Hebbian feedforward connections and anti-Hebbian lateral connections to model neural computation.[28]

Pehlevan's framework provides the mathematical proof for how and why neural populations dynamically adjust their capacity when facing representational limits.[28] The core principle of similarity matching posits that similar inputs should produce similar outputs, which is enforced mathematically by minimizing the difference between the pairwise similarities of inputs and outputs via a Classical Multidimensional Scaling (CMDS) cost function.[28] To achieve true PCA and whitening, this objective function is modified by adding a decorrelating term that penalizes off-diagonal elements of the output covariance matrix, forcing the output channels to decorrelate.[28]

Dropout and Neurogenesis Triggers

In this adaptive PCA model, the number of active output dimensions (or principal neurons) is not fixed. Instead, it is determined dynamically by hard-thresholding the eigenvalues of the input data's covariance matrix.[28]

If the input data has low variance or high redundancy (low signal-to-noise ratio), the network experiences underutilization. Pehlevan's theory predicts that excess neurons will "dropout" or become silent to conserve energy, and their synaptic weights will eventually decay to zero.[28]

Conversely, if the complexity of the streaming input data increases—meaning the number of eigenvalues exceeding the critical threshold grows beyond the number of available principal neurons—the existing network architecture becomes structurally insufficient.[28] Pehlevan's theory explicitly states that when the number of principal components outgrows the current neural capacity, the biological system resolves the bottleneck by adding extra neurons to the circuit via *neurogenesis*.[28]

This theoretical model proves that neurogenesis is not merely a biological phenomenon, but a mathematically optimal algorithmic response to a saturated covariance matrix.[28] When applied to HDC, Pehlevan's framework dictates that an HDC system must monitor the eigenvalue spectrum (or a proxy metric like SNR) of its hypervector bundles. When the signal complexity surpasses the 75 percent capacity threshold, the system must undergo algorithmic neurogenesis, utilizing Sanger's rule to perfectly orthogonalize the newly spawned vectors against the existing saturated bundle.[15]

The Proposed Algorithm: Neurogenesis-Driven Dynamic HDC

Synthesizing the biological principles of Rgl1 progenitor proliferation and sparse updating with the mathematical rigor of Sanger's whitening rule and Pehlevan's adaptive dimensionality yields a novel algorithmic framework for Hyperdimensional Computing. This proposed framework—Neurogenesis-Driven Dynamic HDC—operates in three distinct, sequential phases designed to theoretically circumvent the 75 percent capacity limit.

Phase 1: Capacity Monitoring and the 75% Boundary Trigger

In a standard HDC framework, hypervectors possess a fixed, static dimensionality (e.g., $D =$).[2] As the system encodes streaming data, it continuously bundles new episodic memories or class features into a composite hypervector. Under standard operation, it proceeds until the capacity limit forces catastrophic forgetting.[18]

Under the proposed algorithm, the system employs a lightweight monitoring daemon. This monitor tracks a proxy for the eigenvalue spectrum of the bundle, achieved by measuring the intra-class variance or the cosine similarity of incoming vectors against the bundled state.[21] As the variance expands and the vectors become entangled, the SNR degrades.

When the operational capacity reaches the heuristic 75 percent mark—meaning the separation ratio between intra-class and inter-class vectors has critically narrowed—the monitor generates a "stress" or "saturation" signal.[7] This aligns perfectly with the biological sparse updating hypothesis: the system does not continuously shift its architecture, which would destroy the fixed-size algebra of the VSA. Instead, it waits for the 75 percent boundary condition to be breached, triggering a discrete, structural intervention.[27]

Phase 2: Online Whitening and Decorrelation via Sanger's Rule

Upon hitting the 75 percent threshold, the system temporarily halts standard bundling operations. The current saturated hypervector is fundamentally entangled; it contains high amounts of cross-talk because the historically encoded data points were not perfectly orthogonal.[10]

The system initiates an online whitening phase utilizing an adapted form of Sanger's Generalized Hebbian Algorithm.[15] Treating the saturated hypervector bundle as a stream of input features, Sanger's rule uses anti-Hebbian lateral connections to enforce competition, iteratively extracting the principal components of the bundle.[15] This process isolates the highest-variance (most informative) latent structures from the interference noise.[27]

By projecting the bundled data onto these newly extracted, mutually orthogonal principal

components, the system achieves Signal-Signal Factorization (SSF).[27] The previously saturated hypervector space is "whitened"—decorrelated and variance-equalized. This effectively compresses the existing knowledge, clears the interference crosstalk, and restores the integrity of the representation without losing the underlying data.[15]

Phase 3: Algorithmic Neurogenesis (Dynamic Vector Growth)

Even with successful whitening, a fixed-dimension vector has absolute Shannon information limits. If the data stream continues to introduce novel, complex latent variables, whitening alone will eventually fail because there are simply no available orthogonal dimensions left to exploit.[19]

According to Pehlevan's adaptive dimensionality theory, when the number of significant eigenvalues surpasses the available dimensions, the system must undergo neurogenesis.[28] At this stage, the HDC architecture executes dynamic vector growth.[42] The system mathematically "spawns" a new segment of dimensions (e.g., appending 1,000 new bits to the existing 10,000-dimensional vector, or allocating a completely new auxiliary hypervector to act in tandem).[42]

Crucially, this new vector space cannot be populated with naive random noise. To integrate seamlessly with the existing memory space, the system uses the residual error from the Sanger whitening process to initialize the new vector dimensions.[14] Because Sanger's rule explicitly subtracts the projections of previous principal components, the new "neurons" (hypervector dimensions) are mathematically guaranteed to be strictly orthogonal to the pre-existing, 75 percent-saturated space.[14]

This algorithmic neurogenesis immediately restores the SNR of the system to near 100 percent for new incoming data, circumventing the capacity limit entirely. The system then resumes normal HDC binding and bundling operations in this newly expanded, orthogonalized space until the 75 percent threshold is breached once again, creating an infinite loop of lifelong learning.

Table 1: Comparative Architectural Analysis

System Feature	Traditional Static HDC	Neurogenesis-Driven Dynamic HDC
Dimensionality	Fixed (e.g., strictly $D =$)	Dynamic/Elastic (Grows in discrete orthogonal blocks)
Capacity Limit	Hard mathematical limit; degrades exponentially	Circumvented via adaptive expansion

Updating Paradigm	Continuous bundling regardless of noise	Sparse updating triggered precisely at 75% capacity boundary
Interference Handling	Subject to catastrophic crosstalk and forgetting	Online Whitening via Sanger's GHA prior to expansion
Latent Feature Extraction	Entangled / Superposed	Factorized / Orthogonalized (Maximizes SSF and SNF)
Biological Analogue	Static neural network weights	Adult Neurogenesis (Rgl1 proliferation, BMAL1 trigger)

Hardware Feasibility, Edge AI, and PIM Architectures

The theoretical elegance of a neurogenesis-whitening algorithm must be reconciled with the practical realities of hardware deployment. HDC is highly favored for embedded systems, Edge AI, and Internet of Things (IoT) devices precisely because its core operations (XOR, bitwise addition) map flawlessly to low-power Field-Programmable Gate Arrays (FPGAs) and Processing-In-Memory (PIM) architectures.[7] Standard machine learning algorithms relying on continuous backpropagation and offline EVD/PCA are fundamentally incompatible with these Size, Weight, and Power (SWaP) constrained environments.[7]

Dynamic Dimension Selection in Hardware

Recent advancements in dynamic HDC hardware prove that dynamic vector growth is physically feasible. Frameworks such as *Dynamic-HDC*, *HyperDyn*, and *SHEARer* introduce Dynamic Dimension Selection (DDS), which adaptively masks, prunes, or activates dimensions based on inference requirements.[23]

In a hardware context, algorithmic neurogenesis does not require physically fabricating new circuitry on the fly. Instead, it utilizes the concept of "dark silicon"—pre-fabricated crossbar arrays or memory blocks that are kept powered down to conserve energy.[11] When the 75 percent threshold is reached, the system powers up a dormant block of memory.[11]

Furthermore, the hardware implementation of Sanger's rule is highly efficient because it relies purely on local Hebbian updates (multiply-and-accumulate operations based only on the pre- and post-synaptic activity of connected units). This avoids the massive global data movement

and matrix inversions required by traditional PCA, keeping the energy footprint minimal.[28]

Energy Efficiency and ReRAM Integration

In PIM architectures utilizing memristors or Resistive RAM (ReRAM), standardizing the 75 percent capacity threshold is critical for stability. Analog PIM crossbars suffer from severe non-linear noise and read-disturb vulnerabilities when densely packed.[21] By enforcing a strict 75 percent capacity ceiling, the system avoids the non-linear degradation of analog SNR that occurs at maximum utilization.[21]

The whitening phase, implemented via local anti-Hebbian lateral connections directly within the crossbar, acts as a regularizer. It pushes the analog weights back into a highly discernible, factorized state.[28] The subsequent neurogenesis phase simply redirects new memory allocations to an adjacent, unused ReRAM tile.

Advanced systems like the CubeMind framework demonstrate the viability of this approach in software/hardware co-design. CubeMind utilizes "lazy capacity growth" within its Vector Symbolic Architecture cache (backed by FAISS IndexBinaryFlat).[27] When the cache reaches its capacity limit, the storage size simply doubles dynamically.[27] At a hypervector dimension of 10,240, this highly efficient int8 bipolar caching system can store 490,000 entries using only 5.4GB of memory—vastly outperforming dense float embeddings.[27] Coupling this lazy capacity growth with Sanger whitening would create an edge-capable lifelong learning system with practically infinite bounds.

Implications for Lifelong Learning and Artificial General Intelligence

Circumventing the HDC capacity limit has profound implications for the pursuit of Artificial General Intelligence (AGI) and lifelong machine learning. Traditional deep learning architectures suffer from "catastrophic forgetting"—when trained sequentially on new data distributions, they aggressively overwrite the static synaptic weights that encode previous knowledge.[13]

While Hyperdimensional Computing naturally resists catastrophic forgetting due to its holographic, distributed properties, it ultimately succumbs to capacity saturation.[4] By introducing a neurogenesis-like algorithm, the HDC system successfully mimics the biological hippocampus and olfactory bulb. Old memories are preserved and compressed in the initial vector space, while new, orthogonal dimensions are grown specifically to house novel, unrelated experiences.[25]

Furthermore, the application of Sanger's rule ensures that these new experiences are strictly factorized from the old.[14] As established in the geometry of optimal coding, achieving high Signal-Signal Factorization (SSF) and maintaining pseudo-orthogonality is the primary driver of

systematic generalization.[27] By continuously orthogonalizing the latent space via whitening prior to expansion, the system constructs a perfectly disentangled cognitive map.[27] This allows the HDC agent to rapidly generalize rules learned in one domain to novel concepts encountered in newly expanded dimensions, achieving a level of few-shot generalization and contextual transfer previously unattainable in static associative memories.[27]

Conclusion

The inherent capacity limits of Hyperdimensional Computing, driven by the crosstalk and noise accumulation of vector superposition, present a significant mathematical barrier to the development of robust, lifelong learning architectures. However, by aligning computational design with the biological principles of adult neurogenesis and sparse, boundary-driven updating, these limits can be theoretically and practically circumvented.

Monitoring the hypervector bundle for a 75 percent capacity threshold establishes a highly efficient, sparse trigger mechanism that prevents unrecoverable saturation. Once activated, the application of Sanger's Generalized Hebbian Algorithm acts as an online, streaming whitening filter, decorrelating saturated representations and maximizing information transfer without the need for computationally heavy offline matrix inversions. Following Pehlevan's framework for adaptive dimensionality, when the latent complexity exceeds available dimensions, the algorithm initiates dynamic vector growth. By utilizing the residual error of the whitening process, it spawns new hypervector dimensions that are mathematically guaranteed to be orthogonal to the existing space.

This neurogenesis-driven framework not only bypasses the theoretical Macrow capacity limits but also ensures that the resulting representations remain factorized, disentangled, and optimally structured for downstream generalization. Compatible with the physical constraints of Processing-In-Memory architectures and dynamic hardware masking, this algorithm provides a viable, energy-efficient pathway to evolving Hyperdimensional Computing from a static mapping tool into a limitless, adaptive intelligence.

Works cited

1. A Survey on Hyperdimensional Computing aka Vector Symbolic Architectures, Part II - Redwood Center for Theoretical Neuroscience, accessed March 7, 2026, https://redwood.berkeley.edu/wp-content/uploads/2022/11/2023_CSUR_survey_HDCVSA_part_2.pdf
2. [2112.15475] Shift-Equivariant Similarity-Preserving Hypervector Representations of Sequences - arXiv.org, accessed March 7, 2026, https://arxiv.org/abs/2112.15475
3. Editorial: Brain-inspired Hyperdimensional Computing: Algorithms, models, and architectures - PMC - NIH, accessed March 7, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9775867/
4. EventHD: Robust and efficient hyperdimensional learning with neuromorphic sensor, accessed March 7, 2026,

https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2022.858329/full
5. An Introduction to Hyperdimensional Computing for Robotics | Request PDF - ResearchGate, accessed March 7, 2026, https://www.researchgate.net/publication/335899218_An_Introduction_to_Hyperdimensional_Computing_for_Robotics
6. US20220019441A1 - Circuits, methods, and articles of manufacture for hyper-dimensional computing systems and related applications - Google Patents, accessed March 7, 2026, https://patents.google.com/patent/US20220019441A1/en
7. Frontiers in Edge AI with RISC-V: Hyperdimensional Computing vs. Quantized Neural Networks - DATE 2019, accessed March 7, 2026, https://past.date-conference.com/proceedings-archive/2024/DATA/6039_pdf_upload.pdf
8. SPARSE DISTRIBUTED MEMORY IS A CONTINUAL LEARNER - Xander Davies, accessed March 7, 2026, https://xanderdavies.com/sdm_cl.pdf
9. Lifelong Intelligence Beyond the Edge using Hyperdimensional Computing - arXiv.org, accessed March 7, 2026, https://arxiv.org/html/2403.04759v1
10. Integrating Event-based Dynamic Vision Sensors with Sparse Hyperdimensional Computing - Research Collection, accessed March 7, 2026, https://www.research-collection.ethz.ch/server/api/core/bitstreams/bc4313e3-a387-4c12-aa1d-c7d0b891acf6/content
11. Efficient Biosignal Processing Using Hyperdimensional Computing - IEEE Xplore, accessed March 7, 2026, https://ieeexplore.ieee.org/ielaam/5/8589029/8490896-aam.pdf
12. Adult-neurogenesis allows for representational stability and flexibility in early olfactory system - eLife, accessed March 7, 2026, https://elifesciences.org/reviewed-preprints/107905
13. A neural algorithm for a fundamental computing problem | Request PDF - ResearchGate, accessed March 7, 2026, https://www.researchgate.net/publication/320967614_A_neural_algorithm_for_a_fundamental_computing_problem
14. Reward-Modulated Hebbian Plasticity as Leverage for Partially Embodied Control in Compliant Robotics - Frontiers, accessed March 7, 2026, https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2015.00009/full
15. Hebbian Learning, Principal Component Analysis, and Independent Component Analysis Hebbian Learning The Hebbian Synapse The Hebb, accessed March 7, 2026, https://www.cs.cmu.edu/afs/cs/academic/class/15782-f06/slides/hebbpca.pdf
16. Aspects of hyperdimensional computing for robotics: transfer learning, cloning, extraneous sensors, and network topology - SPIE Digital Library, accessed March 7, 2026, https://www.spiedigitallibrary.org/conference-proceedings-of-spie/11751/117510C/Aspects-of-hyperdimensional-computing-for-robotics--transfer-learning-cloning/10.1117/12.2585772.short

17. Hyperdimensional Computing: Unifying Framework for Enhanced, Effcient and Robust Learning - ScholarWorks, accessed March 7, 2026, https://scholarworks.calstate.edu/downloads/gt54kw090
18. Scalable Edge-Based Hyperdimensional Learning System with Brain-Like Neural Adaptation - NSF PAR, accessed March 7, 2026, https://par.nsf.gov/servlets/purl/10354515
19. Neural Prosthetic with In-sensor Shared Control - EECS, accessed March 7, 2026, https://www2.eecs.berkeley.edu/Pubs/TechRpts/2023/EECS-2023-244.pdf
20. In many classrooms - Acoustical Society of America, accessed March 7, 2026, https://acousticalsociety.org/wp-content/uploads/2022/01/Classroom-Acoustics.pdf
21. A Formal Perspective on Hyperdimensional Computing - eScholarship, accessed March 7, 2026, https://escholarship.org/uc/item/2s9987sh
22. Lineage hierarchies and stochasticity ensure the long-term maintenance of adult neural stem cells - PMC, accessed March 7, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7190328/
23. SHEARer: Highly-Efficient Hyperdimensional Computing by Software-Hardware Enabled Multifold AppRoximation, accessed March 7, 2026, https://cseweb.ucsd.edu/~bkhalegh/papers/ISLPED21-Shearer.pdf
24. Overcoming the challenges of crossbar resistive memory architectures - ResearchGate, accessed March 7, 2026, https://www.researchgate.net/publication/281139462_Overcoming_the_challenges_of_crossbar_resistive_memory_architectures
25. Competition on presynaptic resources enhances the discrimination of interfering memories - PMC, accessed March 7, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10235910/
26. Adult neurogenesis improves spatial information encoding in the mouse hippocampus, accessed March 7, 2026, https://www.biorxiv.org/content/10.1101/2022.11.30.518622v2.full-text
27. s41593-026-02200-8.pdf
28. Optimization theory of Hebbian/anti-Hebbian networks for PCA and whitening - Center for the Neural Basis of Cognition, accessed March 7, 2026, https://www.cnbc.cmu.edu/~tai/nc19journalclubs/pehlevin_pca_hebbian_antihebbian2015.pdf
29. Stimulus whitening improves the efficiency of reverse correlation - PMC, accessed March 7, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10556169/
30. Independent Component Analysis - IME-USP, accessed March 7, 2026, https://www.ime.usp.br/~jstern/miscellanea/seminario/Hyvarinen01.pdf
31. Adult-neurogenesis allows for representational stability ... - bioRxiv.org, accessed March 7, 2026, https://www.biorxiv.org/content/10.1101/2024.07.02.601573v4.full.pdf
32. Stimulus Whitening Improves the Efficiency of Reverse Correlation | bioRxiv, accessed March 7, 2026, https://www.biorxiv.org/content/10.1101/2022.01.25.477706v1.full-text
33. Introduction To The Theory of Neural Computation | PDF - Scribd, accessed

March 7, 2026, https://www.scribd.com/document/385106061/Introduction-to-the-Theory-of-Neural-Computation
34. Michael Biehl - OAPEN Library, accessed March 7, 2026, https://library.oapen.org/bitstream/handle/20.500.12657/93936/The%20Shallow%20and%20the%20Deep.pdf?sequence=1&isAllowed=y
35. (PDF) A Review of Findings from Neuroscience and Cognitive Psychology as Possible Inspiration for the Path to Artificial General Intelligence - ResearchGate, accessed March 7, 2026, https://www.researchgate.net/publication/377472846_A_Review_of_Findings_from_Neuroscience_and_Cognitive_Psychology_as_Possible_Inspiration_for_the_Path_to_Artificial_General_Intelligence
36. csinva.github.io/_notes/neuro/comp_neuro.md at master · csinva/csinva.github.io · GitHub, accessed March 7, 2026, https://github.com/csinva/csinva.github.io/blob/master/_notes/neuro/comp_neuro.md
37. The generation and dynamic control of neural activity sequences - eDiss, accessed March 7, 2026, https://ediss.uni-goettingen.de/bitstream/handle/11858/15845/Lehr_PhD_Thesis.pdf?sequence=1&isAllowed=y
38. Real-time Continual Learning on Intel Loihi 2 - arXiv.org, accessed March 7, 2026, https://arxiv.org/html/2511.01553v1
39. Beyond gradients: Noise correlations control Hebbian plasticity to shape credit assignment, accessed March 7, 2026, https://www.biorxiv.org/content/10.1101/2021.11.19.466943v1.full-text
40. Unveiling Principles of Neural Computations: From Biological to Artificial Intelligence, and Back - Kreiman Lab, accessed March 7, 2026, https://klab.tch.harvard.edu/publications/PDFs/gk8125.pdf
41. A Lightweight Embedded PPG-Based Authentication System for Wearable Devices via Hyperdimensional Computing - MDPI, accessed March 7, 2026, https://www.mdpi.com/1999-4893/19/1/83
42. Integrating Branching and Pruning for Efficient Hyperdimensional Computing | Request PDF, accessed March 7, 2026, https://www.researchgate.net/publication/384975020_Integrating_Branching_and_Pruning_for_Efficient_Hyperdimensional_Computing
43. Locality-based Encoder and Model Quantization for Efficient Hyper-Dimensional Computing - NSF PAR, accessed March 7, 2026, https://par.nsf.gov/servlets/purl/10253141
44. comp neuro - Chandan Singh, accessed March 7, 2026, https://csinva.io/notes/neuro/comp_neuro.html
45. HyperDyn: Dynamic Dimensional Masking for Efficient Hyper-Dimensional Computing | Request PDF - ResearchGate, accessed March 7, 2026, https://www.researchgate.net/publication/391962549_HyperDyn_Dynamic_Dimensional_Masking_for_Efficient_Hyper-Dimensional_Computing
46. Neurips 2023 compressed - yobihome, accessed March 7, 2026,

https://yobibyte.github.io/neurips2023_compressed.html